Nguyễn Hữu Chí Bảo - N25DECE046

Thứ ngày tháng năm

P1 2,1
P2 2,5
P3 2,5
P4 2,5
~~5,85~~
9,60

Problem 1: Singly Linked List - In-Place Reversal & Pointer Tracing

- Task 1: Write a function / algorithm in pseudocode to reverse the singly linked list, include three pointers (prev, curr, next) to complete your function
- Task 2: Draw a diagram or step-by-step pointer trace table showing the position and values of the three pointers (prev, curr, next) after each iteration until the list is fully reversed.

Problem 2: Circular Linked List - Insert at Head

- Given circular linked list head → [10] → [20] → [30] → (points back to head)
- Task 1: Write pseudo code for the function Insert Head (Head, New value) to insert a new node containing new value at the front of the circular linked list while maintaining its circular structure.
- Task 2: Trace the execution stop when inserting new value = 5 into the given list.

Problem 3: Double Linked List.

Given a doubly linked list containing integer

- Task 1: Write pseudocode for Insert Before (head, target, new value)

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Task 2: List the exact order of pointer assignments required and briefly to justify why this specific sequence avoids memory leak or broken links.

Task 3: Explicitly handle the following two edge case in code:

1. target is located at the head node.
2. target does not exist in the list.

Problem 4: Sorted Linked list

Given a singly linked list sorted in ascending order

head → [2] → [5] → [8] → NULL

Task 1: Write a function SortedInsert

Task 2 Construct a step-by-step trace the algorithm for iteration and conditional evaluation when insert if val = 6 into the list.

Bài làm.

~~Task~~

2.1 * Problem 1. Task 1: Pseudocode.

1.1 Reverse (head):

prev = NULL

curr = head

next = NULL

while curr ≠ NULL

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```
next = curr.next
curr.next = prev
prev = curr
curr = next
return prev
```

1.0 Task 2. Trace table (list 10 → 25 → 40 → 15)

Linked Struct -0,25

Step	prev	curr	next	Action
Init	NULL	10	–	–
1	10	25	25	10.next = NULL
2	25	40	40	25.next = 10
3	40	15	15	40.next = 25
4	15	NULL	NULL	15.next = 40

Then stop when curr = NULL. Return prev, which is 15, so result is 15 → 40 → 25 → 10 → NULL.

2.5 Problem 2. Task 1: Pseudocode.

1.25 Insert Head (head, new value)

```
node = new Node(new value)
if(head = NULL)
    node.next = node
    return node
tail = head
```


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```
while (tail.next ≠ head:)
    tail = tail.next.
node next = head
tail.next = node
return node
```

1.25 Task 2. Trace, newValue = 5

1 Create node[5].

2 The list not Empty. Set tail = 10.

3. 10.next = 20 ≠ head, so tail = 20

4. 20.next = 30 ≠ head, so tail = 30

5 30.next = head (10), so stop.

6. Set node(5).next = 10

7. Set 30.next = 5

8 Set head = 5.

Result: 5 → 10 → 20 → 30 → (back to 5)

2.5 * Problem 3: Task 1+3: Pseudo code with edge cases.

1.75 Insert Before (head, target, new Value):

```
curr = head.
while (curr ≠ NULL and curr.data ≠ target.)
    curr = curr.next.
if (curr = NULL)
    return head
```


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node = new Node (new value)
if (curr = head)
    node.prev = NULL
    node.next = head
    head.prev = node
    return node.
node.prev = curr.prev
node.next = curr
curr.prev.next = node
curr.prev = node
return head.

0,75 Task 2

1. node.prev = curr.prev.
2. node.next = curr
3. curr.prev.next = node
4. curr.prev = node

2,5 * Problem 4: Task 1: pseudo code.

1,25 Sorted Insert (head, val):
    node = new Node (val)
    if head = NULL or val ≤ head.data:
        node.next = head

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```
return node.
curr = head
while (curr.next ≠ NULL and curr.next.data < val)
    curr = curr.next
node.next = curr.next
curr.next = node
return head.
```

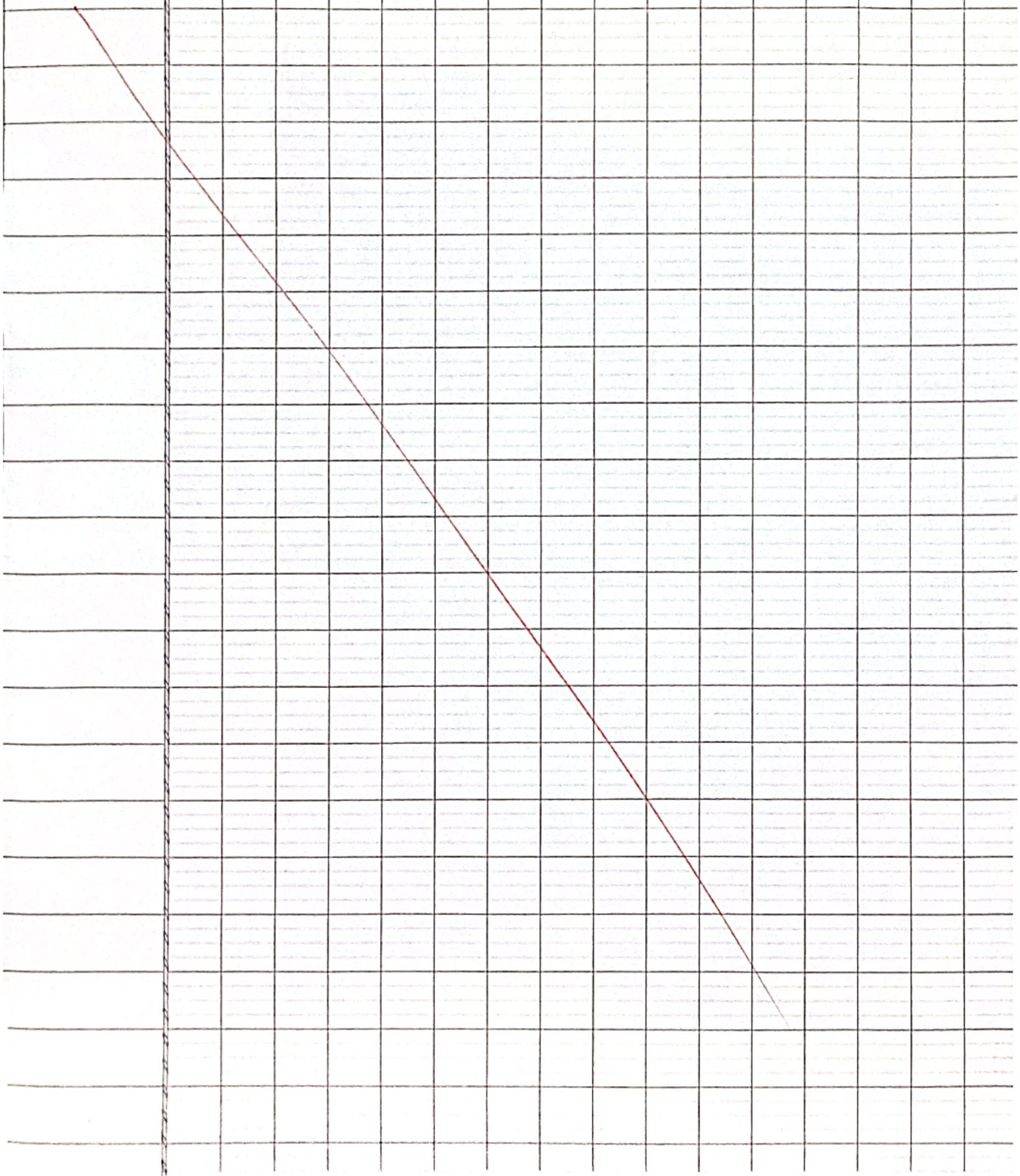
1.25 Task 2. Trace, val = 6 (list 2→5→8)

Step	curr	curr.next	Condition curr.next != NULL and curr.next.data < 6	Action
check head	2	5	6 <= 2 ? False	Not insert at head
1	2	5	5 < 6 → True	curr = 5
2	5	8	8 < 6 → False	Stop loop
Insert	5	8	–	node(6).next = 8 5.next = 6

Result 2 → 5 → 6 → 8 → NULL

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